

Project Executable Files

Date	4 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes (crop_stats.json)
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

/OptiCrop-Project
|-- 1. Brainstorming & Ideation
|-- 2. Requirement Analysis
|-- 3. Project Design Phase
|-- 4. Project Planning Phase
|-- 5. Project Development Phase
|   |-- app.py (Flask Server Backend Routing & API)
|   |-- train_model.py (ML Pipeline Classifier Training)
|   |-- model.pkl (Serialized Random Forest Classifier Model)
|   |-- crop_stats.json (Aggregated Ideal Nutrient Ranges Database)
|   |-- Crop_recommendation.csv (Raw Agricultural Dataset)
|   |-- static/ (CSS Styles & UI Asset Files)
|   |-- templates/ (Jinja2 HTML Interface Views)
|-- 6. Project Testing
|-- 7. Project Documentation
`-- 8. Project Demonstration
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://127.0.0.1:5001
Login Credentials (Demo)	No login credentials required (Evaluation dashboards are public)
Platform / Hosting Provider	Local Host Server (Python / Flask)
Repository Link	https://github.com/burlaprudhviraj123/OptiCrop-Smart-Agricultural-Production-Optimization-Engine-Smartbridge
Demo Video Link	https://drive.google.com/file/d/1h9dzmD2WpUbYlc4oxG437qkKIBDaNR8i/view?usp=share_link

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- Python 3.12 (or Python 3.10+) installed.
- Modern web browser (Chrome, Edge, Firefox, or Safari).

Step 1: Open the terminal and navigate to the project development directory:

```
cd "/Users/burlapudhviraj/Downloads/Smartbridge/OptiCrop-Project/5. Project Development Phase"
```

Step 2: Activate the pre-configured Python virtual environment:

```
source venv/bin/activate
```

Step 3: (Optional) To retrain the machine learning model and calculate crop statistics:

```
python train_model.py
```

Step 4: Start the local Flask web development server:

```
python app.py
```

Step 5: Open your web browser and navigate to the local address:

```
http://127.0.0.1:5001
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Local server dependencies setup	Virtual environment (venv) is pre-configured for instant loading.
2	Chart.js offline loading limitations	Visual dashboard requires active internet connection for Chart.js CDN.
3	Model retraining overhead	Trained model is cached as model.pkl; training script can re-run in seconds.